

#if statement

a=33

b=200

if b>a:

print("b is greater than a")

#else

a=200

b=33

if b>a:

print("b is a greater than a")

else:

print("b is not greater than a")

a=200

b=20

if b>a:

print("b is a greater than a")

elif a==b:

print("a and b are equal")

else:

print("b is not greater than a")

amount=2000

if amount <1000:

discount=amount*0.05

```
print("discount",discount)

elif amount<5000:

    discount=amount*0.10

    print("discount",discount)

else:

    discount=amount*0.15

    print("discount",discount)

    print("net payable",amount-discount)
```

#nested if statement

```
var=100

if var<200:

    print("expression value is less than 200")

    if var==150:

        print("which is 150")

    elif var==100:

        print("which is 100")

    elif var==50:

        print("which is 50")

    elif var<50:

        print("expression value is less than 50")

    else:

        print("could not find true expression")

print("good bye!")
```

```
var=185

if var<200:

    print("expression value is less than 200")

    if (var>150 and var<170):

        print("which is greater than 150 less than 170")

    elif (var==100 or var>180):

        print("which is equal to 100 or greater than 180")

    elif var==50:

        print("which is 50")

    elif var<50:

        print("expression value is less than 50")

    else:

        print("could not find true expression")


print("good bye!")
```

#while statement

```
count=0

while (count<9):

    print("the count is:",count)

    count=count+1

print("good bye!")
```

```
count=0

while count<7:

    print (count,"is less than 7")

    count=count+1
```

else:

print("loop end")

#for loop statement

range(0,5)

[0,1,2,3,4]

for var in list (range(5)):

print(var)

for i in range(0,30,3):

print(i)

for i in range(0,50,5):

print(i)

#loop control statement

for i in range (1,6):

if i==3:

break

print(i)

for i in range (1,6):

if i==3:

continue

print(i)

```
for i in range(1,4):  
    pass  
    print("loop completed")
```

```
x=10  
if x>5:  
    pass  
else:  
    print("less than 5")  
print("done")
```

#python format() method

```
name="john"  
print("hello, %s!" % name)
```

```
name="john"  
age=23  
print("%s is %d years old." % (name,age))
```

```
txt1="my name is{fname},I'm{age}".format(fname="john",age=36)
```

```
txt2="my name is{0},I'm {1}".format("john",36)
```

```
txt3="my name is{},I'm {}".format("john",36)
```

```
print(txt1)
```

```
print(txt2)
```

```
print(txt3)
```

```
str="java"
```

```
str2="C#"
```

```
str3="{ } and { } both are programming languages".format(str,str2)
```

```
print(str3)
```

```
num1=input('enter first number:')
```

```
num2=input('enter second number:')
```

```
sum=float(num1)+float(num2)
```

```
min=float(num1)-float(num2)
```

```
mul=float(num1)*float(num2)
```

```
div=float(num1)/float(num2)
```

```
print('the sum of {0} and {1} is {2}'.format(num1,num2,sum))
```

```
print('the subtraction of {0} and {1} is {2}'.format(num1,num2,min))
```

```
print('the multiplication of {0} and {1} is {2}'.format(num1,num2,mul))
```

```
print('the division of {0} and {1} is {2}'.format(num1,num2,div))
```

```
x=input('enter value of x:')
```

```
y=input('enter value of y:')
```

```
temp=x
```

```
x=y
```

```
y=temp
```

```
print ('the value of x after swapping :{}'.format(x))
```

```
print ('the value of y after swapping :{}'.format(y))
```

```
num=int(input("enter a number:"))
```

```
if(num % 2)==0:
```

```
    print("{} is even number".format(num))
```

```
else:
```

```
    print("{} is odd number".format(num))
```

output

b is greater than a

b is not greater than a

b is not greater than a

discount 200.0

expression value is less than 200

which is 100

good bye!

expression value is less than 200

which is equal to 100 or greater than 180

good bye!

the count is: 0

the count is: 1

the count is: 2

the count is: 3

the count is: 4

the count is: 5

the count is: 6

the count is: 7

the count is: 8

good bye!

0 is less than 7

1 is less than 7

2 is less than 7

3 is less than 7

4 is less than 7

5 is less than 7

6 is less than 7

loop end

0

1

2

3

4

0

3

6

9

12

15

18

21

24

27

0

5

10

15

20

25

30

35

40

45

1

2

1

2

4

5

loop completed

loop completed

loop completed

done

hello, john!

john is 23 years old.

my name isjohn,I'm36

my name isjohn,I'm 36

my name isjohn,I'm 36

java and C# both are programming languages

enter first number:10

enter second number:2

the sum of 10 and 2 is 12.0

the subtraction of 10 and 2 is 8.0

the multiplication of 10 and 2 is 20.0

the division of 10 and 2 is 5.0

enter value of x:10

enter value of y:5

the value of x after swapping :5

the value of y after swapping :10

enter a number:2

2 is even number